

Tribenzalol Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-106/B

Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

Deviations observed during the campaign were investigated and closed with no impact to product quality. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection.

No changes to the approved analytical procedures are proposed in this submission. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. The control strategy links critical quality attributes to critical process parameters identified during development. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection.

Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.

Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

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Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Cleaning verification swab results were below the health-based exposure limits derived for the product. Deviations observed during the campaign were investigated and closed with no impact to product quality. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data.

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No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Cleaning verification swab results were below the health-based exposure limits derived for the product.

Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

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The control strategy links critical quality attributes to critical process parameters identified during development. No changes to the approved analytical procedures are proposed in this submission. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions.

P.4 Control of Excipients

For the commercial formulation F-106/B, Parteck M200 (Mannitol, diluent) is used.

Selection Justification

The selection of Parteck M200 is justified by improved flowability during high-speed compression observed during development studies.

Grade selection and control follow the principles of IPEC Excipient Qualification Guide and FDA Guidance for Industry: Immediate Release Solid Oral Dosage Forms.

Commercial Control Strategy — Excipients

Component	Function	Grade per current specification
Mannitol	diluent	Pardeck M200
Reference		SPEC-EX-7754 Rev 04

Analytical Method Validation Summary

The assay and related-substances procedures employ reversed-phase high performance liquid chromatography with ultraviolet detection. Validation demonstrated acceptable specificity, linearity, accuracy, precision, and robustness. Forced degradation studies confirmed the stability-indicating capability of the related-substances method.

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Statistical evaluation of batch release data demonstrates process capability well within specification limits. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. The control strategy links critical quality attributes to critical process parameters identified during development. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Cleaning verification swab results were below the health-based exposure limits derived for the product.

Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

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